

Elena Zhang

Atlanta, GA 30313 | (404) 422-6106 | elzhang.0219@gmail.com | [linkedin.com/in/elena-zhang-19/](https://www.linkedin.com/in/elena-zhang-19/) | github.com/elzhang19 | F-1 visa holder

Objective

Computer Engineering major at Georgia Tech with hands-on experience in embedded systems, PCB design, and applied research. Skilled in leadership, problem solving, and presentation skills.

Education

Georgia Institute of Technology | Atlanta, GA

June 2025 – Present

Bachelor of Science in Computer Engineering, GPA 4.00

Expected Graduation, May 2029

- Relevant coursework: Digital System Design (in progress), Data Structures and Algorithms (in progress), Introduction to Object-Oriented Programming (Java)

University of New South Wales | Sydney, Australia

February 2025 – April 2025

Bachelor of Science in Computer Science

- Co-op scholarship recipient (~\$20k p.a.) selected by representatives from TikTok, Atlassian, and Insurance Australia Group

Skills

Programming: Python, Java, R, C++

Embedded and hardware: Arduino, PCB Design (Altium), basic circuit analysis, breadboard prototyping, soldering, UART communication

Tools: Git/GitHub, VS Code, IntelliJ, Jupyter, Google Colab, Overleaf, Microsoft Office Suite

Communication: Teaching, mentorship, leadership, presentations

Experience

Vertically Integrated Projects (VIP) Mechatronics and Motivation team | Atlanta, GA

January 2026 – present

Undergraduate Researcher

- Developing wearable electronics that integrate EMG sensors and haptic actuators for real-time affective state monitoring in K-12 mathematics learning contexts.
- Onboarding onto the haptics PCB and sensor firmware subteams, contributing to system architecture and schematic review.

Bio-Medical Informatics and Bioimaging Laboratory (Bio-MIBLab) | Atlanta, GA

August 2025 – December 2025

Undergraduate Researcher

- Evaluated and compared LLM reasoning schemes (Tree, Graph, Buffer of Thoughts) for EEG-based obstructive sleep apnea classification.
- Gained proficiency running LLMs on Georgia Tech's high-performance computing (HPC) PACE-ICE cluster.

North Shore Tutors | Sydney, Australia

November 2024 – June 2025

Academic Tutor

- Delivered one-on-one tutoring lessons to 3 high school students in mathematics and chemistry; witnessed improvement in test results by one letter grade average.
- Curated lesson plans based on course outcomes of the Higher School Certificate (HSC) and Cambridge IGCSE curricula.

Projects

Arduino Morse Code Translator

September 2025 – November 2025

ECE1100 Discovery Project

- Designed embedded system using Arduino Uno to encode/decode Morse code via LEDs, piezo buzzer, and push-button inputs.
- Implemented bidirectional translation logic and timing control in firmware (C++) and validated functionality using Tinkercad circuit simulation.

Activities

RoboNav (Georgia Tech Mars rover robotics team) | Electrical sub-team

August 2025 – Present

- Designing and testing an autonomous line-following rover.
- Training in circuit analysis, breadboard prototyping, Arduino-based firmware development, soldering, crimping, UART communication, and hardware testing using multimeters.